



HUST

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HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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The background of the entire image is a dark blue field filled with a pattern of red dots. These dots are arranged in a way that they form a large, stylized, abstract shape that resembles a heart or a pair of hands reaching towards each other. The dots are of varying sizes and are distributed across the entire frame, creating a textured, digital effect.

SOICT

School of Information and Communication Technology

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Software Quality Assurance and Testing

3. Introduction to Software Testing

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Contents

- Definition and Motivation of Software Testing
- Basic terms of Software Testing
- Test Activities
- Test Levels



What is software testing?

- IEEE defines software testing as:
 - "A process of analyzing a software item to detect the **difference between existing and required conditions** (defects/errors/bugs) and to evaluate the features of the software item"
- Note that...
 - ...one needs to know the **required conditions**
 - ...one needs to be able to observe the **existing conditions**



Software Testing is a process

- Evaluating a program or a system
 - Verify that the product is right
 - Validate that the right product is developed
 - Evaluate the product quality
 - Find defects



Testing takes time

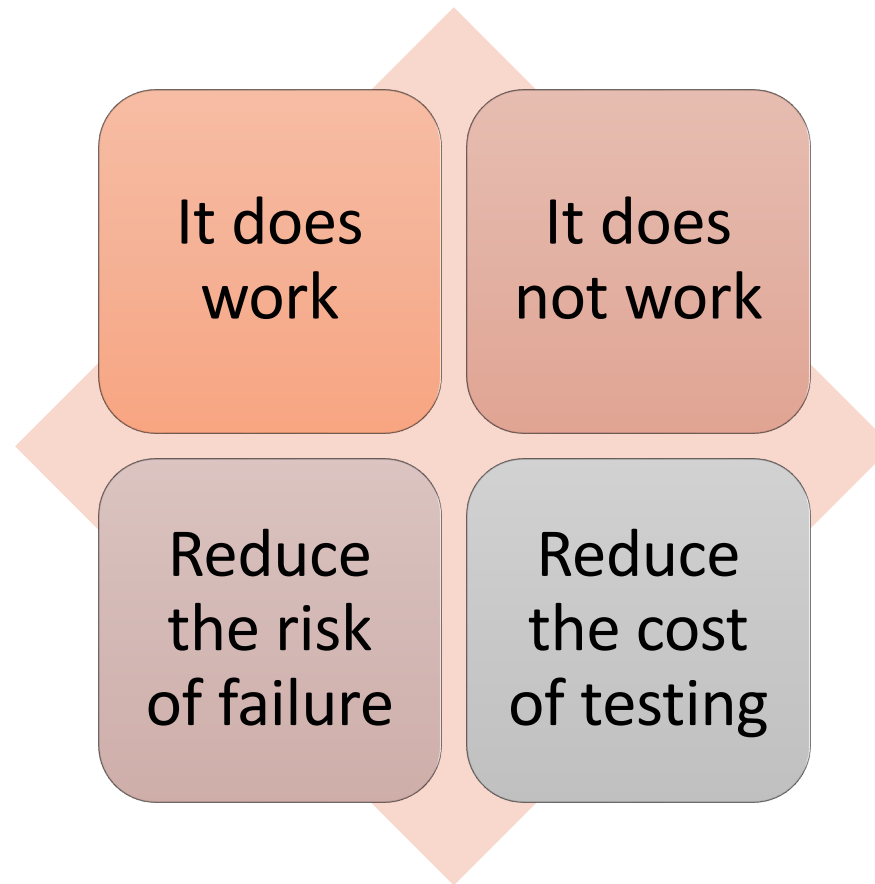
“Reports estimate that regression testing consumes as much as 80% of the testing budget and can consume up to 50% of the cost of the software maintenance”

–Chittimalli and Harrold, 2009

Chittimalli, P, & Harrold, M 2009, 'Recomputing coverage information to assist regression testing', IEEE Transactions On Software Engineering, 35, 4, pp. 452-469



Objectives of Testing



Test case

Test case là gì/Cách viết?

- Simply, a pair of <input, expected outcome>
- Example: a square root function of nonnegative numbers
- Two kinds of testing function
 - stateless function/system
 - example: compiler
 - state-oriented system
 - example: ATM machine to with

TB₁: < 0, 0 >,
TB₂: < 25, 5 >,
TB₃: < 40, 6.3245553 >,
TB₄: < 100.5, 10.024968 >.

TS₁: < check balance, \$500.00 >, < withdraw, "amount?" >,
< \$200.00, "\$200.00" >, < check balance, \$300.00 >



Expected Outcome

- An **outcome** of program execution may include
 - Single values produced by the program
 - State changes
 - Sequence or set of values which must be interpreted together for the outcome to be validated
- The concept of **oracle** in test designing: any entity (program, process, body of data...) producing the expected outcome of a particular test
- Expected outcome should be computed when designing test case
 - Without executing program but using program's requirements/specifications



Complete Testing

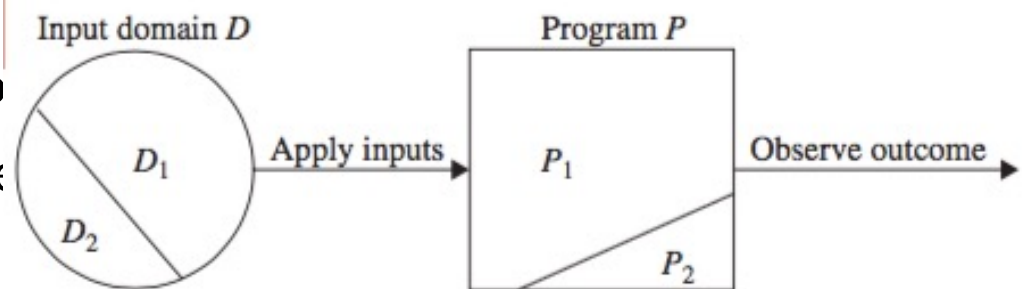
Hàn tất kiểm thử

- Complete Testing means:
 - There are no **undiscovered** faults at the end of the test phase
- It is impossible to complete testing for most of the software systems
 - **Domain** of possible inputs of a program is too large to be completely used in testing
 - **valid** and **invalid** values of data input
 - **Design issues** may be too complex to completely test
 - Impossible to create all possible **execution environments** of the system



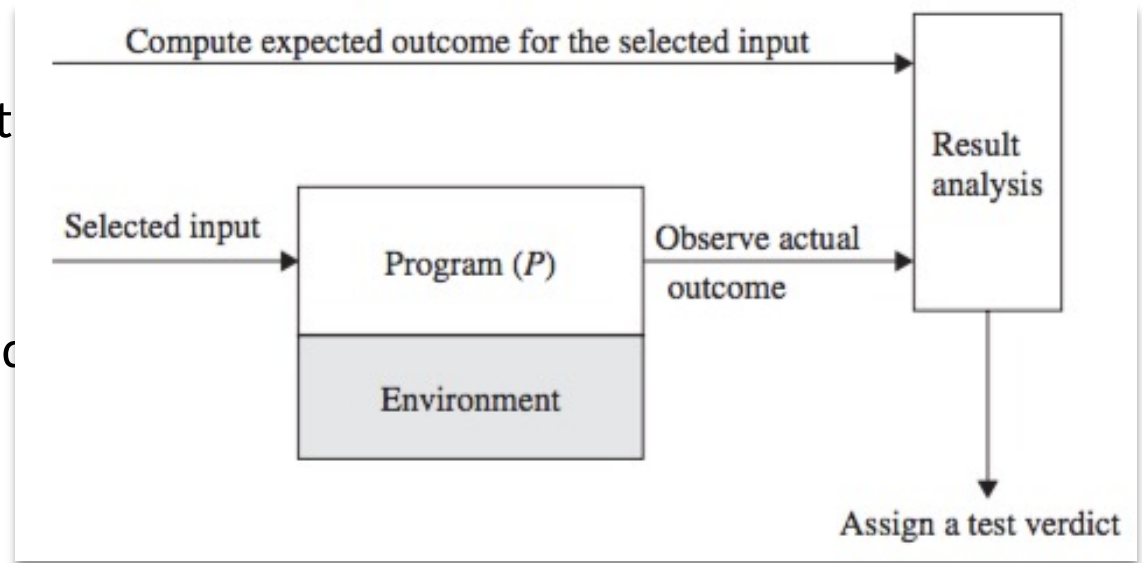
Centre Issue in Testing

- Select a subset of the input domain
- Let D be the input of the program P
 - Select subset $D_1 \subset D$ to test P : D_1 ex (behavior of P)
 - We said that we were attempting to deduce properties of P on selected inputs D_1



Testing Activities

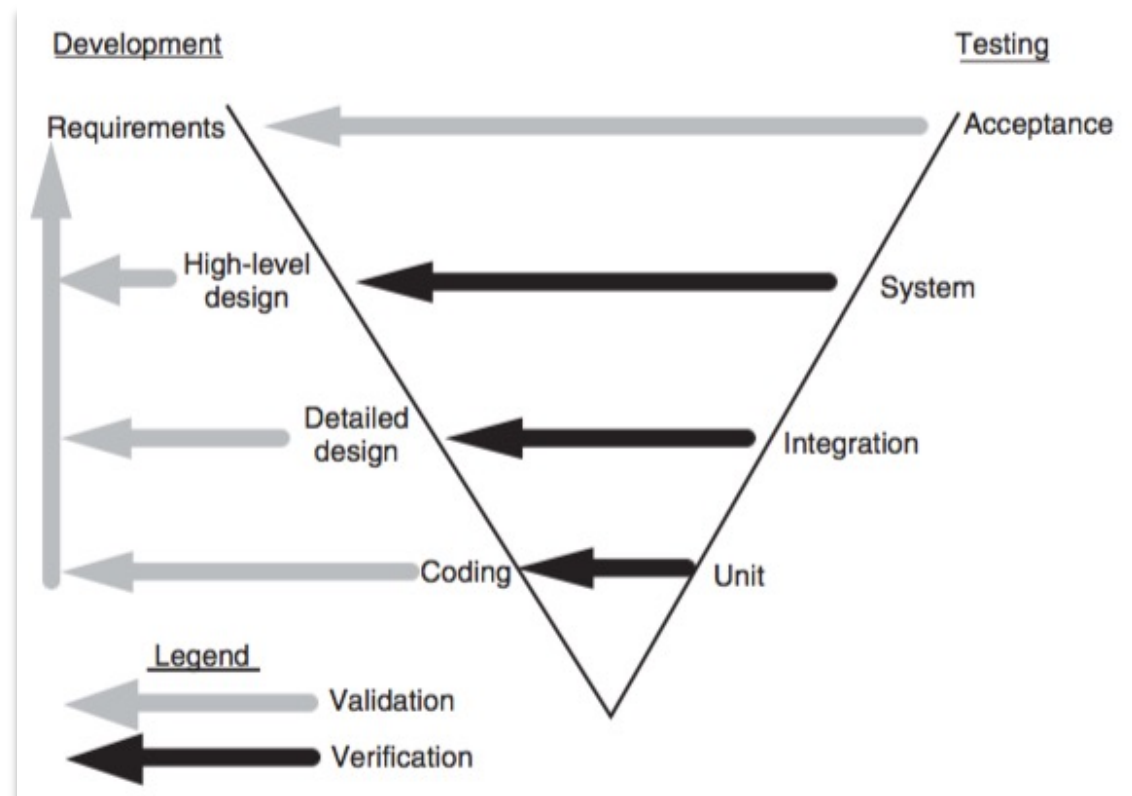
- Identifying an objective to be tested
- Select inputs
- Compute the expected outcome
- Setup the execution environment of t
- Execute the program
- Analyze the test result
- Writing test report if test case is failed



Test Levels

Mức độ kiểm thử?

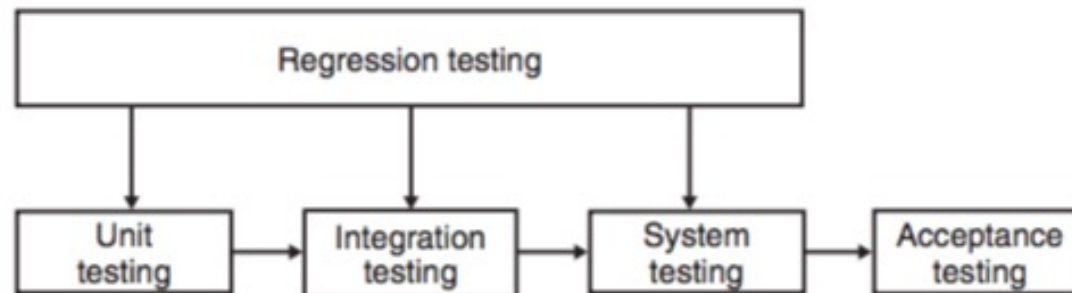
- Testing is performed at different levels involving the whole system or parts of it throughout the life cycle of a software product
- Four stages of testing
 - Unit test
 - Integration test
 - System test
 - Acceptance test



Regression Test

Kiểm thử hồi quy

- Subphase of unit/integration/system testing
- Performed whenever a component is modified
- To ensure that no new faults are introduced because of performed modifications
- No new tests are considered but tests are selected, prioritized and executed



Acceptance Test

Kiểm thử chấp nhận

- Objective of acceptance testing is to measure the quality of the product rather than searching for the defects
- Based on expectations of customers
- 2 kinds of acceptance testing
 - User acceptance testing (UAT)
 - Business acceptance testing (BAT)



Sources of Information for Test Case

Nguồn thông tin để thiết kế và kiểm thử?

- Objective of designing test cases: generate **effective** tests at a **lower cost**
- Sources of Information:
 - Requirements and functional specifications
 - Source code
 - Input and Output domains
 - Operational Profile
 - Fault Model



Requirements and Functional Specifications

- The process of software development begins by capturing user needs
- The number of user needs identified at the beginning depends on the specific life-cycle model
 - ex: waterfall vs agile
- Test case designing is performed based on software requirements
- Requirements can be specified in different manners
 - Plaintext, equations, figures, flowcharts...
 - Use cases, entity-relationship diagrams, class diagrams
 - Using formal language and notation as finite-state machine...



Source Code

- Requirement specifications describe the intended behavior of a system
- Source Code describes the actual behavior of the system
- Test cases must be designed based on the program



Input and Output domains

- Input domain
 - Valid data and Invalid data
- Some values in the input domain may have special meanings and should be treated separately
- Special values in the input and output domains of a program should be considered while testing the program



Operational Profile

- **Operation/Usage profile** is a quantitative characterization of how a system will be used
- It is important to test a system by considering the ways it will actually be used
- To guide test engineers in selecting test cases using samples of system usage
- Test inputs are assigned a probability distribution or profile according to their occurrences in actual operation

Fault Model

- Encountered faults are a excellent source of information in designing new test cases
- Different classes of faults: initialization faults, logic faults, interface faults
- 3 types of fault-based testing:
 - Error guessing
 - Fault seeding
 - Mutation testing



Test case designing techniques

- White Box Testing
 - Structural testing techniques
 - Examining source code with a focus on control flow and data flow
 - Applied to individual units of a program
- Black Box Testing
 - Functional testing techniques
 - Examining functional specifications of testing modules
 - Applied to both an entire system and the individual program units
- These strategies are used by different groups of people at different times during a software life cycle
 - Ex: programmers use both to test their own code while quality assurance engineers apply the idea of functional testing



Test Planning and Design

- The purpose of system test planning is to get ready and organized for test execution
- Test plan provides a framework, scope, details of resource needed, effort required, schedule of activities and a budget
- Test design is a critical phase of software testing
 - System requirements are critically studied
 - System features to be tested are identified
 - Objectives and detailed behaviors of test cases are defined
- Test-Driven Development (TDD): test cases are designed and implemented before the production code is written
 - key practice of modern agile software development processes

Monitoring and Measuring test execution

- Motivation: to explore the progress of testing and reveal the quality level of the system
- Test execution metrics are categorized into 2 classes
 - Metrics for monitoring test execution
 - Metrics for monitoring defects
- Quantitative measurements
 - Number of defects detected by test cases
 - Number of test cases designed/executed per day
 - Number of defects found by the customers that were not found by the test engineers

Summary

- Students should understand principle terms of software testing
- Have a general understand of V-model
- Understand different phase/activities of testing process





Software Quality Assurance and Testing

(end of lecture)

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